

Username:

CS441

2025-09-17

Name:

Discrete Structures for CS

hw_3

Problem 1. Let $P(x, y)$ be the statement “Student x has taken class y ,” where the domain for x consists of all students in your class and y consists of all computer science courses at your school. Express each of these quantifications in english.

- a) $\exists x \exists y P(x, y)$
- b) $\exists x \forall y P(x, y)$
- c) $\forall x \exists y P(x, y)$
- d) $\exists y \forall x P(x, y)$
- e) $\forall y \exists x P(x, y)$
- f) $\forall x \forall y P(x, y)$

Answer.

Problem 2. Let $Q(x, y)$ be the statement “Student x has been a contestant on quiz show y .” Express such of these sentences in terms of $Q(x, y)$, quantifiers, and logical connectives, where the domain for x consists of all students at your school and for y consists of all quiz shows on television.

- a) There is a student at your school who has been a contestant on a television quiz show.
- b) No student at your school has ever been a contestant on a television quiz show.
- c) There is a student at your school who has been a contestant on *Jeopardy!* and on *Wheel of Fortune*.
- d) Every television quiz show has had a student from your school as a contestant.
- e) At least two students from your school have been contestants on *Jeopardy!*.

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Problem 3. Express each of these statements using predicates, quantifiers, logical connectives, and mathematical operators where the domain consists of all integers.

- a) The product of two negative integers is positive.
- b) The average of two positive integers is positive.
- c) The difference of two negative integers is not necessarily negative.
- d) The absolute value of the sum of two integers does not exceed the sum of the absolute values of these integers.

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Problem 4. Find a common domain for the variables x , y , and z for which the statement $\forall x \forall y ((x \neq y) \rightarrow \forall z ((z = x) \vee (z = y)))$ is true and another domain for which it is false.

Answer.

Problem 5. What rule of inference is used in each of these arguments?

- a) Kangaroos live in Australia and are marsupials.
Therefore, kangaroos are marsupials.
- b) It is either hotter than 100 degrees today or the pollution is dangerous.
It is less than 100 degrees outside today.
Therefore, the pollution is dangerous.
- c) Linda is an excellent swimmer.
If Linda is an excellent swimmer, then she can work as a lifeguard.
Therefore, Linda can work as a lifeguard.
- d) Steve will work at a computer company this summer.
Therefore, this summer Steve will work at a computer company or he will be a beach bum.
- e) If I work all night on this homework, then I can answer all the exercises.
If I answer all the exercises, I will understand the material.
Therefore, if I work all night on this homework, then I will understand the material.

Answer.

Problem 6. For each of these arguments, explain which rules of inference are used for each step.

- a) “Linda, a student in this class, owns a red convertible. Everyone who owns a red convertible has gotten at least one speeding ticket. Therefore, someone in this class has gotten a speeding ticket.”
- b) “Each of five roommates, Melissa, Aaron, Ralph, Veneesha, and Keeshawn, has taken a course in discrete mathematics. Every student who has taken a course in discrete mathematics can take a course in algorithms. Therefore, all five roommates can take a course in algorithms next year.”
- c) “All movies produced by John Sayles are wonderful. John Sayles produced a movie about coal miners. Therefore, there is a wonderful movie about coal miners.”
- d) “There is someone in this class who has been to France. Everyone who goes to France visits the Louvre. Therefore, someone in this class has visited the Louvre.”

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Problem 7. For each of these arguments determine whether the argument is correct or incorrect and explain why.

- a) Everyone enrolled in the university has lived in a dormitory. Mia has never lived in a dormitory. Therefore, Mia is not enrolled in the university.
- b) A convertible car is fun to drive. Isaac's car is not a convertible. Therefore, Isaac's car is not fun to drive.
- c) Quincy likes all action movies. Quincy likes the movie *Eight Men Out*. Therefore, *Eight Men Out* is an action movie.
- d) All lobstermen set at least a dozen traps. Hamilton is a lobsterman. Therefore Hamilton sets at least a dozen traps.

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Problem 8. What is wrong with this statement? Let $S(x, y)$ be “ x is shorter than y .” Given the premise $\exists s S(s, \text{Max})$, it follows that $S(\text{Max}, \text{Max})$. Then by existential generalization it follows that it follows that $\exists x S(x, x)$, so that someone is shorter than himself.

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Problem 9. Use rules of inference to show that if $\forall x (P(x) \vee Q(x))$ and $\forall x ((\neg P(x) \wedge Q(x)) \rightarrow R(x))$ are true, then $\forall x (\neg R(x) \rightarrow P(x))$ is also true, where the domains of all quantifiers are the same.

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Challenge 1. Show that $\forall x P(x) \vee \forall x Q(x)$ and $\forall x \forall y (P(x) \vee Q(y))$, where all quantifiers have the same nonempty domain, are logically equivalent. (The new variable y is used to combine the quantifications correctly.)

Answer.